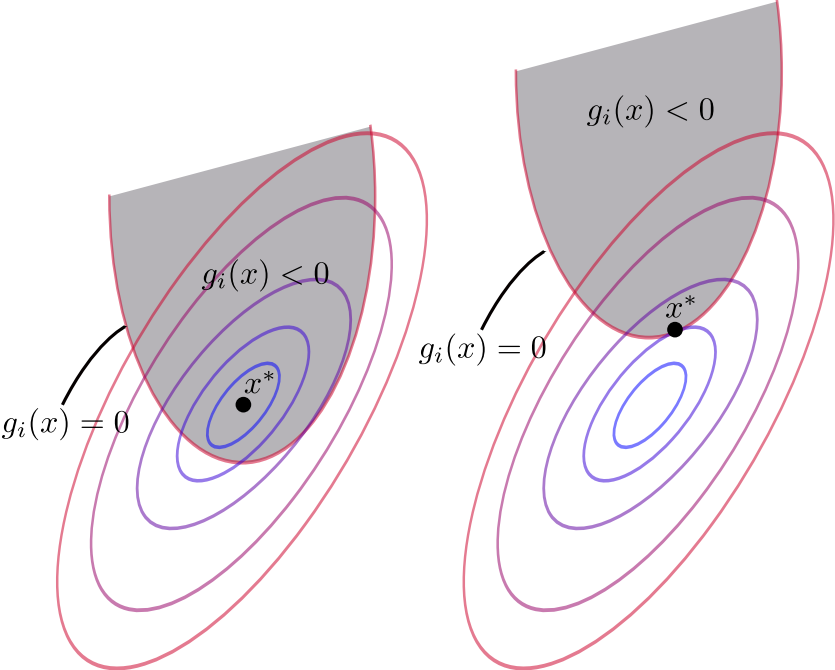
$g_i(x) < 0$
x^*
$g_i(x) = 0$
$g_i(x) < 0$
x^*
$g_i(x) = 0$